Wide Area Networks (WANs)

What is a WAN?

Wide area networks covers a large geographical area, they could be worldwide. Because of this, devices may be provided by telecoms companies like phone lines and satellites.

What is the Internet?

The Internet is the biggest WAN in the world. The internet is a massive network of networks. In other words, it is a ginormous collection of connected computers.

Is the Internet and the World Wide Web the same?

Although these two terms are used interchangeably, they are in fact very different. As explained above, the Internet is a ginormous collection of connected computers. The World Wide Web however, is just one application / use of the Internet. The WWW is the web of websites, providing the sharing of information. Other applications of the Internet are E-mail and FTP downloading.

Web Hosting

A host is a computer on a network which offers to serve users from another location, using the internet. Web hosting, in particular, is a service provided by various companies. These companies 'host' websites on their servers so that people around the world can access them 24/7. They will often charge a small fee for the rental of hard drive space, site security and back up services. It is of course possible for anyone (with a little knowhow) to host their own websites on their home PC, but for the price and risk, it is often more cost effective and less risky to pay for a specialist company to do this for you.

Web Hosting and DNS

In order to connect to various webhosts, so that we can access the websites we need, we use a system called DNS. In order to understand how DNS works we first need to understand a few other important acronyms.

IP Address

This means INTERNET PROTOCAL ADDRESS. It is a unique number given to every computer on the internet – no two computers can have the same address (e.g. 324.45.321.23). It's just like a postal address – used to identify a house – no two houses have the same address! A device's IP address will not necessarily remain the same each time it joins a network or goes on-line.

ISP

This means INTERNET SERVICE PROVIDER. This is simply the company who provide you with your internet connection.

URL

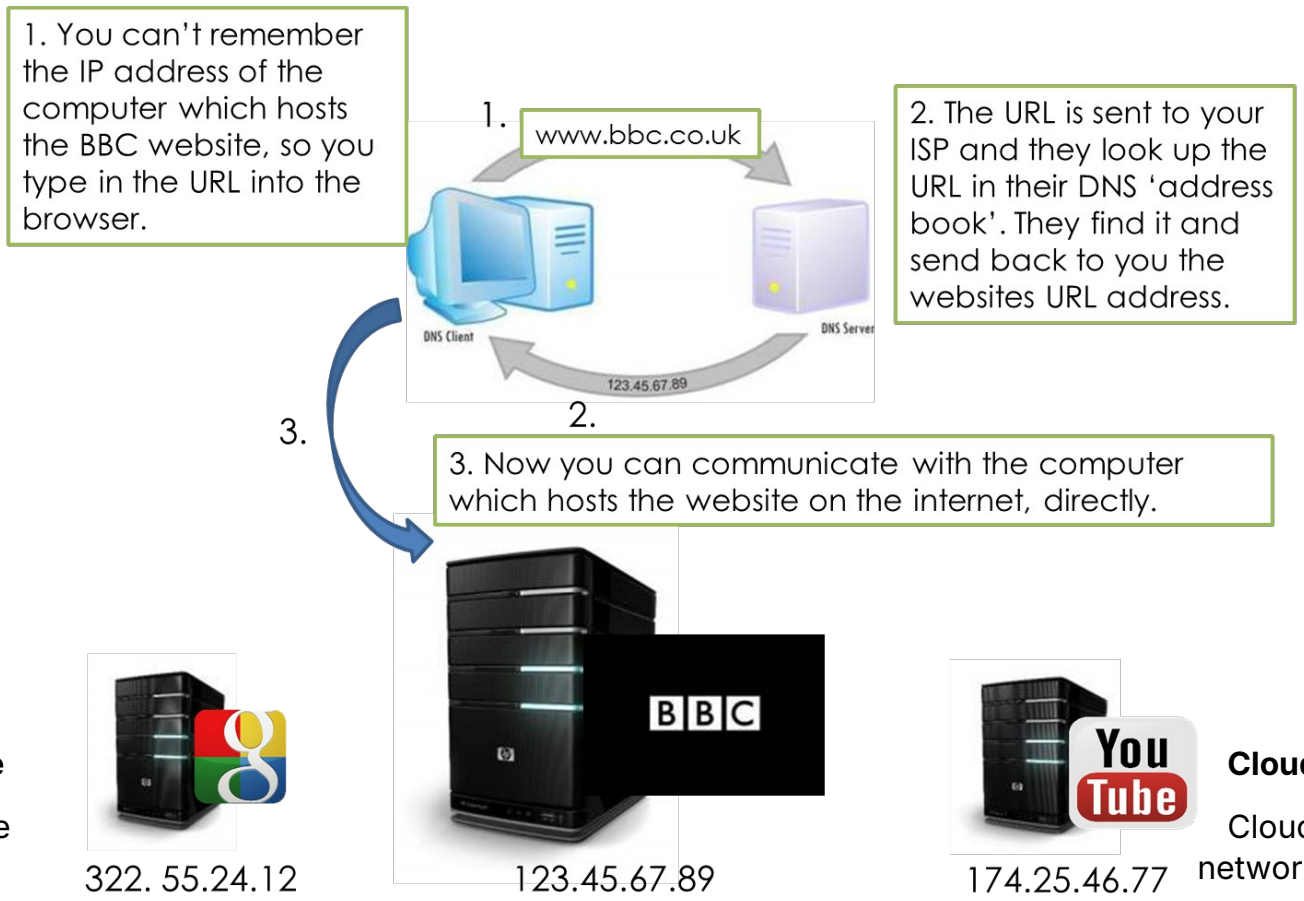
This means UNIFORM RESOURCE LOCATOR. This is simply a fancy name for a web address, such as "http://www.bbc.co.uk" or "http://www.google.com".

DNS

Computer Science UK Membership Site Licence: Do not share outside your centre.

This means DOMAIN NAME SYSTEM. This is the system used to find the computer which hosts the website you are looking for.

How does DNS work?



The **Cloud**

'The Cloud' is a network of servers hosted on the internet which offer a range of services to store and process data. In particular, 'Cloud Computing' relates to online services provided by numerous companies.

Examples of cloud computing are:

- Office 365 / GoogleDocs
- Cloud Storage (GoogleDrive, OneDrive, DropBox)
- Netflix / Spotify

Why would a company use such a service as office365?

- They do not need to 'buy' and install the software.
- Any 'connected' computer can access the service.
- There is no need to upgrade the software.
- Collaboration can occur with multiple people working on the same document at once.
- Work is automatically saved / backed up.

This generally all leads to lower costs.

But...

- Sensitive company data may be stored in another country which may not adhere to the same data laws as the country the company is in.
- Completely reliant on the network / internet.

Virtual Networks

We should by now be happy with the idea that LANs are a network of computers over a small geographical location and WANs are a network of computers over a wider geographical area. For both types of network, they are created through the connection of physical hardware.

A virtual network is one which uses software to subdivide a physical network (LAN or WAN) into smaller ones.

Virtual Networks and LANs (VLANs)/

For a simple virtual network, software will re-divide a LOCAL AREA NETWORK into a series of smaller networks so that (for example) groups of workers can communicate and share information separately from other members of a company. Because it is software which enables the network to partitions (hardware remains as it was) it is knows as a VIRTUAL network.

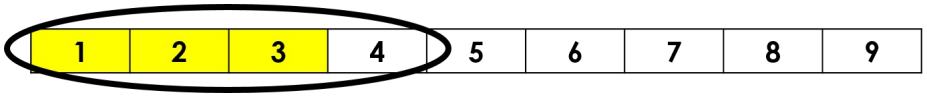
Virtual Private Networks (VPN) and WANs

Software can also re-divide a WIDE AREA NETWORK into a series of smaller networks so that (for example) groups of workers spread across the country / world can communicate and share information separately from other members of a company. As the internet is used by all, these networks will need added security and this is done through encryption. Due to this they are given the name PRIVATE NETWORKS, but of course, because they are created through the use of software, they are known as Virtual Private Networks (VPN).

Questions *(The question zone you choose must either match your target grade or be higher!)*

Computer Science UK Membership Site Licence: Do not share outside your centre.

Question Zone 1-3



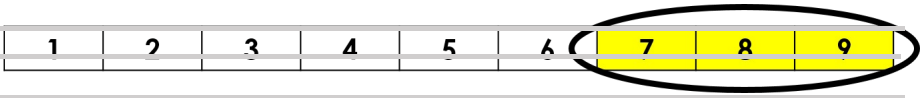
1. Write a definition of a WAN. [2]
2. Describe, using examples, what is meant by the acronyms IP address, ISP and URL. [6]
3. Describe, using an example, what is meant by the acronym DNS. [5]

Question Zone 4-6



1. Describe, using an example, what is meant by the acronym DNS. [5]
2. Explain, using examples, what is meant by the term 'The Cloud'? [2]
3. Explain what a virtual network is, providing examples of different types of virtual network. [5]

Question Zone 7-9



1. Explain what a virtual network is, providing examples of different types of virtual network. [5]
2. Explain the difference between the Internet and the World Wide Web, using examples to support your answer. [4]
3. Discuss the advantages and disadvantages of an organisation using cloud computing services. [8]

Stick answer sheet here

State/Identify/Give/Name: Simply label a diagram, fill out a table or write a few words
Describe: Describing is 'saying what you see' (E.G.: A computer will have a CPU, Primary and Secondary storage etc)
Explain: Explaining is 'saying WHY/HOW something is like that'. (E.G.: A computer will have a CPU so that it can process all of the data the computer needs to perform a range of tasks. Primary and Secondary storage is needed because...)
Discuss: Discussing is 'looking at two sides of an issue, weighing up the two views and giving a conclusion'. Often these require a mini essay answer. (E.G.: New technology could be seen as being bad for the environment because..., but on the other hand, new technology has led to... In conclusion I believe that...)
Describe/Explain/Discuss **using examples**: Finally, if you are asked to give examples in any of these types of questions – YOU MUST **GIVE EXAMPLES**.

Computer Science Teacher Membership Site Licence. Do not share.

Checklist:

- Date and title, clearly presented
- Spelling & grammar checked
- Question numbers in the margin
- Handwriting neat & legible
- Punctuation / Capital letters

Keywords / Key Terms:

WAN: *Wide Area Network – A network that covers are large geographical area and whose equipment (phone lines / satellites) is usually owed by third party telecommunication companies often supply.*

Cloud Computing: A term used to describe online services and applications

URL*: Uniform Resource Locator – a fancy name for the web address of a website.*

Computer Science UK Membership Site Licence: Do not share outside your centre.

Stick answer sheet here

State/Identify/Give/Name: Simply label a diagram, fill out a table or write a few words
Describe: Describing is 'saying what you see' (E.G.: A computer will have a CPU, Primary and Secondary storage etc)
Explain: Explaining is 'saying WHY/HOW something is like that'. (E.G.: A computer will have a CPU so that it can process all of the data the computer needs to perform a range of tasks. Primary and Secondary storage is needed because...)
Discuss: Discussing is 'looking at two sides of an issue, weighing up the two views and giving a conclusion'. Often these require a mini essay answer. (E.G.: New technology could be seen as being bad for the environment because..., but on the other hand, new technology has led to... In conclusion I believe that...)
Describe/Explain/Discuss **using examples**: Finally, if you are asked to give examples in any of these types of questions – YOU MUST **GIVE EXAMPLES**!

Computer Science UK Membership Site Licence: Do not share outside your centre.

Reflections **Score:** / **Percentage:** % **Grade:** **Progress** On / Above / Below

What Went

- I demonstrated a good level of understanding.
- I responded to the command words effectively.
- My answers were detailed / were written in depth.
- My work was well presented / legible.
- My answers effectively incorporated technical terminology.
- My responses were well structured / organised.
- My revision strategy was effective as I showed depth of understanding in my answers.
- My answers contained enough points / examples / explanations to achieve the marks available.

Even Better

- My answers need to be more accurate.
- I must respond correctly to the command words.
- My answers need more detail / greater depth.
- I must take greater care over my work / write neatly.
- I must incorporate key terminology into my answers.
- I must better organise my answers to improve its clarity.
- I need to improve my revision strategy as I did not demonstrate a depth of understanding in my answers.
- My answers didn't contain enough points / examples / explanations to achieve the marks available.

Further thoughts:

_____

_____

_____

_____

Computer Science UK Membership Site Licence: Do not share outside your centre.